Reinforcement Learning

An Introduction

Course Content

1. Introduction to Reinforcement Learning
2. Markov Decision Processes (MDPs)
3. Policy and Value Functions
4. Dynamic Programming (DP) for RL
5. Model-Free methods
6. Value Function Approximation
7. Policy-Gradient and Actor-Critic Methods
8. Deep RL
9. TP Project

Modern Applications

RLHF

OpenAI (when it was still open)

Aligning LLMs to human preferences

How to transform a Raw GPT model into an *Instruct* One ?

Prompt

Explain the moon landing to a 6 year old in a few sentences.

Completion

GPT-3

```
Explain the theory of gravity to a 6 year old.

Explain the theory of relativity to a 6 year old in a few sentences.

Explain the big bang theory to a 6 year old.

Explain evolution to a 6 year old.
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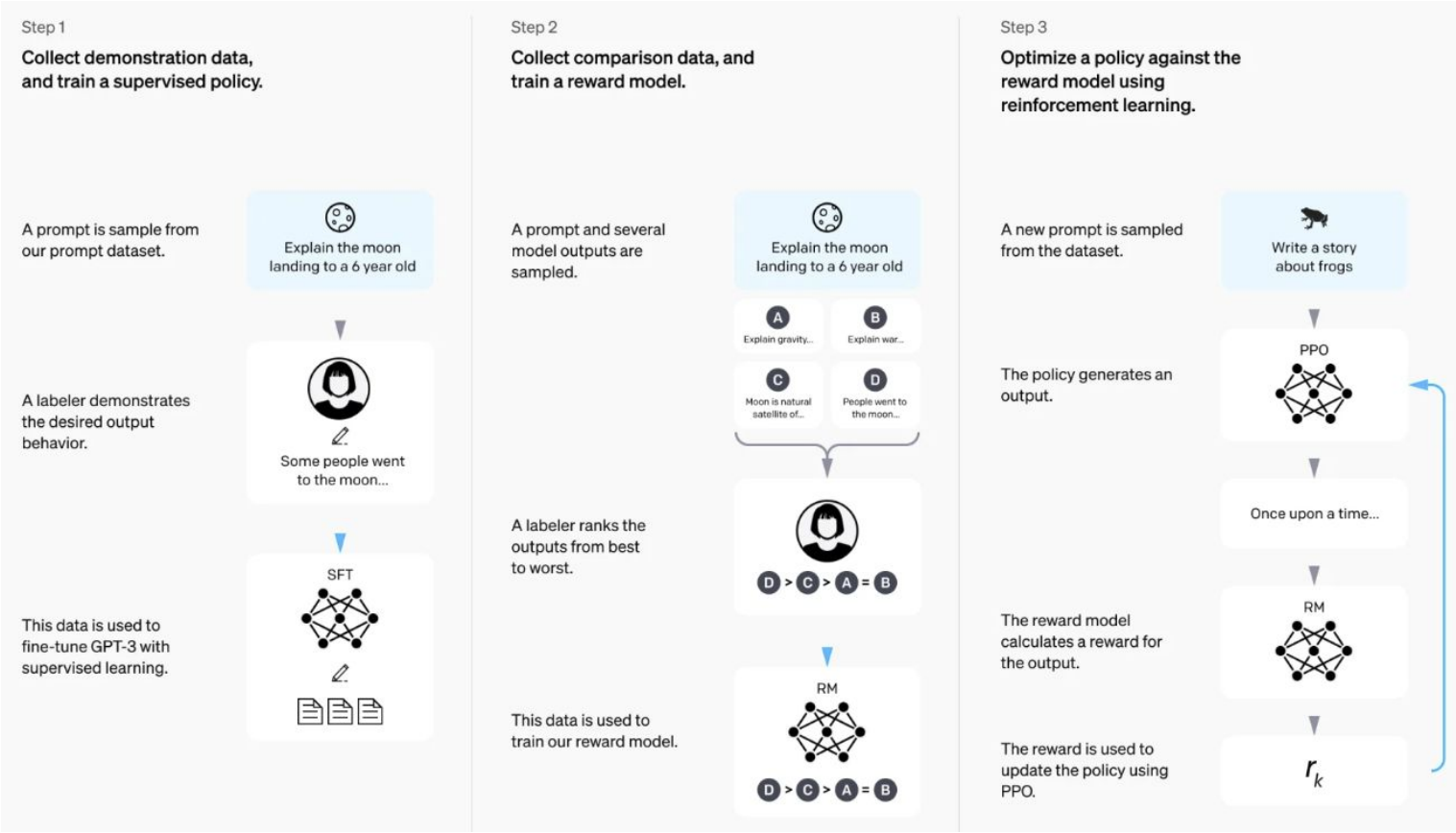
InstructGPT

```
People went to the moon, and they took pictures of what they saw, and sent them back to the earth so we could all see them.
```

Test Raw GPT-2 here !

Method - Using PPO !



More Details in the OpenAI [original RLHF post]

GRPO

When DeepSeek entered the Game

GRPO - why the hype ?

This made a little storm in 2024

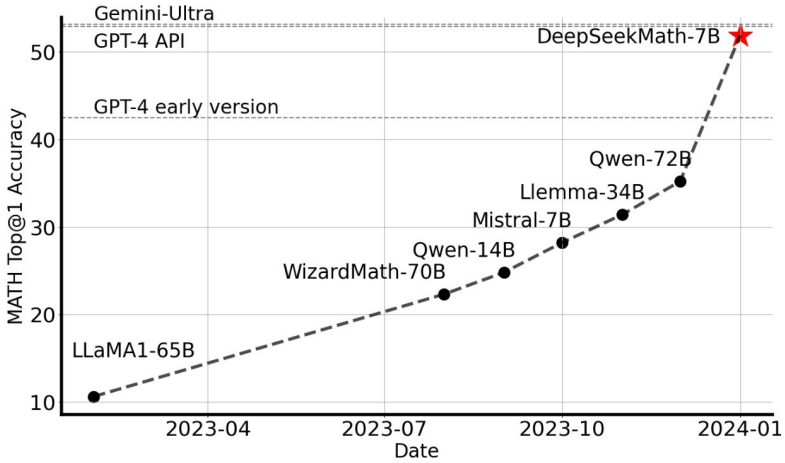


Figure 1 | Top1 accuracy of open-source models on the competition-level MATH benchmark (Hendrycks et al., 2021) without the use of external toolkits and voting techniques.

DeepSeekMath: Pushing the Limits of Mathematical Reasoning in Open Language Models

GRPO - Principle

Simple steps:

- For each question, the model **generates several possible answers**.
- Each answer is scored using a reward model (e.g., giving a high score for a correct math answer).
- The average score of all answers for that question is calculated.
- The model compares each answer's score to this average to see how good it is (this is called the "**Advantage**").
- The model then updates itself to favor answers with higher advantages, ensuring it doesn't change too much at once to stay stable.

This process repeats, helping the model get better over time. A surprising detail is how it uses the group average as a baseline, which reduces the need for extra memory while still improving performance.

DeepSeekMath: Pushing the Limits of Mathematical Reasoning in Open Language Models

GRPO - Algo

This made a little storm back in

Algorithm 1 Iterative Group Relative Policy Optimization

Input initial policy model $\pi_{\theta_{\text{init}}}$; reward models r_φ; task prompts $\mathcal{D}$; hyperparameters ε, β, μ

```
policy model π_θ ← π_θ_init
for iteration = 1, ..., I do
    reference model π_ref ← π_θ
    for step = 1, ..., M do
        Sample a batch D_b from D
        Update the old policy model π_θ_old ← π_θ
        Sample G outputs {o_i}_{i=1}^G ~ π_θ_old(· | q) for each question q ∈ D_b
        Compute rewards {r_i}_{i=1}^G for each sampled output o_i by running r_φ
        Compute Â_{i,t} for the t-th token of o_i through group relative advantage estimation.
        for GRPO iteration = 1, ..., μ do
            Update the policy model π_θ by maximizing the GRPO objective (Equation 21)
    Update r_φ through continuous training using a replay mechanism.
```

Output π_θ

DeepSeekMath: Pushing the Limits of Mathematical Reasoning in Open Language Models

GRPO - Comparison

How to transform a Raw GPT model into an *Instruct* One ?

Compared to PPO, GRPO **eliminated the need of a Critic** !
Specially well tailored for LLMs

Algo	Value based	Advantage Estimation	Stability Mechanism	Memory Usage
PPO	✅	Uses value function for baseline	Clipped surrogate objective	High (due to value function)
TRPO	✅	Uses value function, trust region constraint	Hessian-based trust region	High
REINFORCE	❌	No baseline, high variance	None	Low
GRPO	❌	Group mean as baseline, reduces variance	KL divergence in loss function	Low

More Details in [this good blogPost]

Drone Racing

nature
UZH
UZH

Now it’s your turn..

Any Questions ?

Contact us !

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References

[1] David Silver, Lectures on Reinforcement Learning, 2015

[2] Reinforcement Learning and Advanced Deep Learning (Sorbonne) - Olivier Sigaud

[3] Sutton, R. S. and Barto, A. G. (2018), Reinforcement Learning: An Introduction (Second edition). MIT Press

Olivier Sigaud Youtube Channel